INCH-POUND

MIL-DTL-3970/16C
28 April 2008
SUPERSEDING
MIL-W-3970/16B
4 April 1977

DETAIL SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE H-PLANE MITERED CORNER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design
after 18 September 1998.

Requirements for acquiring the waveguide assemblies described herein shall consist of this document and MIL-DTL-3970.

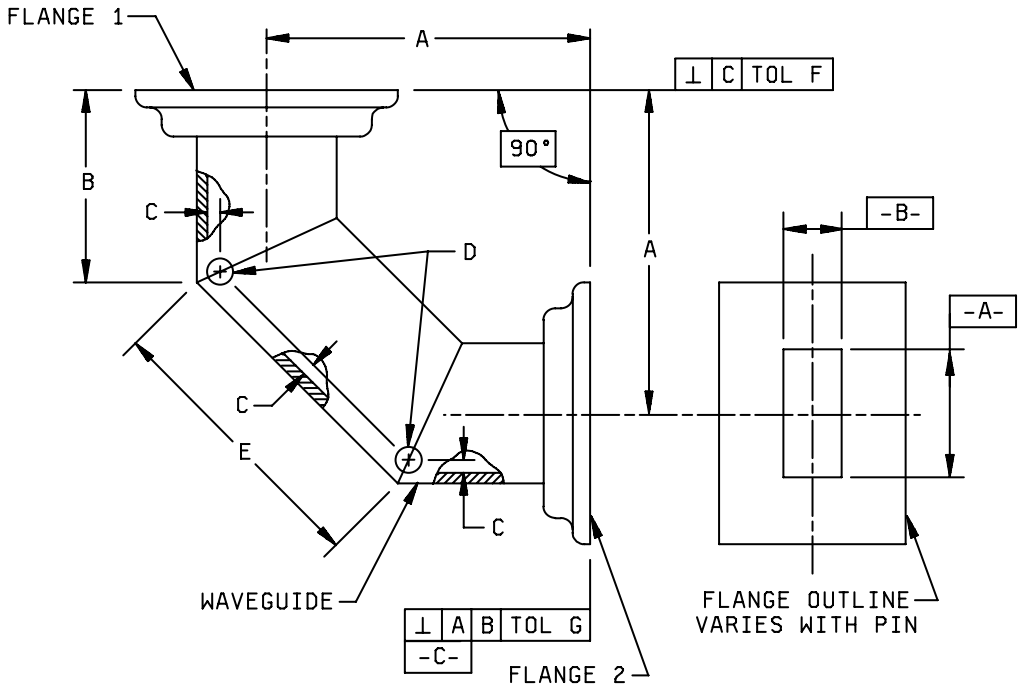

FIGURE 1. Assembly configuration.

AMSC N/A

FSC 5985

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TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash no.	Wave-guide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressur-ization lb_f/in^2 gauge	Dimensions 1/ 2/ 3/ A	B	C	Pin dia B	E	F	G
				GHz	dB									
001	1-043	56-001	61-002	2.60 - 3.95	.05	1.09	15	4.225 (107.32) ±.020 (±0.51)	2.625 (66.68) ±.015 (±0.38)	.278 (7.06) ±.005 (±0.13)	.394 (10.01) ±.005 (±0.13)	4.384 (111.35) ±.020 (±0.51)	.060 (1.52)	.012 (0.30)
002	1-055	57-002	62-002	3.95 - 5.85	.05	1.08	30	2.924 (74.57) ±.020 (±0.51)	1.875 (47.63) ±.015 (±0.38)	.184 (4.67) ±.005 (±0.13)	.260 (6.60) ±.005 (±0.13)	2.898 (73.61) ±.020 (±0.51)	.041 (1.04)	.008 (0.20)
003	1-067	55-001	60-001	5.85 - 8.20	.05	1.08	30	2.139 (54.33) ±.015 (±0.38)	1.375 (34.93) ±.010 (±0.25)	.136 (3.45) ±.003 (±0.08)	.190 (4.83) ±.003 (±0.08)	2.141 (54.38) ±.015 (±0.38)	.032 (0.81)	.006 (0.15)
004	1-073	53-002	59-007	7.05 - 10.0	.06	1.08	30	1.869 (47.47) ±.015 (±0.38)	1.250 (31.75) ±.010 (±0.25)	.110 (2.79) ±.002 (±0.05)	.156 (3.96) ±.002 (±0.05)	1.760 (44.70) ±.015 (±0.38)	.024 (0.61)	.005 (0.13)
005	1-079	53-001	59-006	8.20 - 12.40	.06	1.10	30	1.370 (34.80) ±.010 (±0.25)	0.875 (22.23) ±.010 (±0.25)	.088 (2.24) ±.002 (±0.05)	.125 (3.18) ±.002 (±0.05)	1.410 (35.81) ±.010 (±0.25)	.019 (0.48)	.004 (0.10)
006	1-102	54-001	59-003	18.0 - 26.50	.11	1.06	30	.775 (19.69) ±.010 (±0.25)	.550 (13.97) ±.010 (±0.25)	.040 (1.02) ±.002 (±0.05)	.0580 (1.47) ±.0005 (±0.01)	.672 (17.07) ±.010 (±0.25)	.011 (0.28)	.002 (0.05)
007	1-106	54-001	59-003	18.0 - 26.50	.11	1.06	30	.775 (19.69) ±.010 (±0.25)	.550 (13.97) ±.010 (±0.25)	.040 (1.02) ±.002 (±0.05)	.0580 (1.47) ±.0005 (±0.01)	.672 (17.07) ±.010 (±0.25)	.011 (0.28)	.002 (0.05)
008	1-077	53-003	59-008	8.20 - 12.40	.06	1.10	30	1.370 (34.80) ±.010 (±0.25)	.875 (22.23) ±.010 (±0.25)	.088 (2.24) ±.002 (±0.05)	.125 (3.18) ±.002 (±0.05)	1.410 (35.81) ±.010 (±0.25)	.019 (0.48)	.004 (0.10)

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash no.	Wave-guide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressur-ization lb_f/in^2 gauge	Dimensions 1/ 2/ 3/ A	B	C	Pin dia B	E	F	G
				GHz	dB									
009	1-071	53-004	59-009	7.05 - 10.0	.06	1.08	30	1.869 (47.47) ±.015 (±0.38)	1.250 (31.75) ±.010 (±0.25)	.110 (2.79) ±.002 (±0.05)	.156 (3.96) ±.002 (±0.05)	1.760 (44.70) ±.015 (±0.38)	.024 (0.61)	.005 (0.13)
010	1-041	56-002	61-001	2.60 - 3.95	.05	1.09	15	4.225 (107.32) ±.020 (±0.51)	2.626 (66.68) ±.015 (±0.38)	.278 (7.60) ±.005 (±0.13)	.394 (10.01) ±.005 (±0.13)	4.384 (111.35) ±.020 (±0.51)	.060 (1.52)	.012 (0.30)
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.10	30	1.091 (27.71) ±.010 (±0.25)	.750 (19.05) ±.010 (±0.25)	.060 (1.52) ±.002 (±0.05)	.087 (2.21) ±.001 (±0.03)	.978 (24.84) ±.010 (±0.25)	.015 (0.38)	.003 (0.08)
012	1-053	57-001	62-001	3.95 - 5.85	.05	1.08	30	2.924 (74.57) ±.020 (±0.51)	1.875 (47.63) ±.015 (±0.38)	.184 (4.67) ±.005 (±0.13)	.260 (6.60) ±.005 (±0.13)	2.898 (73.61) ±.020 (±0.51)	.041 (1.04)	.008 (0.20)
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.12	30	.645 (16.38) ±.010 (±0.25)	.500 (12.70) ±.010 (±0.25)	.027 (0.69) ±.002 (±0.05)	.0390 (0.99) ±.0005 (±0.01)	.459 (11.66) ±.010 (±0.25)	.009 (0.23)	.001 (0.03)
014	1-065	55-002	60-002	5.85 - 8.20	.05	1.08	30	2.139 (54.33) ±.015 (±0.38)	1.375 (34.93) ±.010 (±0.25)	.136 (3.45) ±.003 (±0.08)	.190 (4.83) ±.003 (±0.08)	2.141 (54.38) ±.015 (±0.38)	.032 (0.81)	.006 (0.15)
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.10	30	1.091 (27.71) ±.010 (±0.25)	.750 (19.05) ±.010 (±0.25)	.060 (1.52) ±.002 (±0.05)	.087 (2.21) ±.001 (±0.03)	.978 (24.84) ±.010 (±0.25)	.015 (0.38)	.003 (0.08)
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.06	30	.775 (19.69) ±.010 (±0.25)	.550 (13.97) ±.010 (±0.25)	.040 (1.02) ±.002 (±0.05)	.0580 (1.47) ±.0005 (±0.01)	.672 (17.07) ±.010 (±0.25)	.011 (0.28)	.002 (0.05)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part or Identifying Number (PIN): M3970/16-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

PIN M3970/16-	Type designation	PIN M3970/16-	Type designation
001	UG-844/U	009	UG-852/U
002	UG-845/U	010	UG-853/U
003	UG-846/U	011	UG-854/U
004	UG-847/U	012	UG-855/U
005	UG-848/U	013	UG-856/U
006	UG-849/U	014	UG-857/U
007	UG-850/U	015	UG-858/U
008	UG-851/U	016	UG-859/U

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. This specification sheet references only MIL-DTL-3970.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2008-035)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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